

Compilation to Bosonic Qudit Systems

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Motivation

Many quantum systems naturally support more than two levels, allowing them to function as qudits. Using expanded state spaces may enable more efficient ways to encode information and implement quantum circuits

Our Goal:
Develop methods of decomposing Target Unitaries into gate sequences executable on bosonic qudit systems

Core Building Blocks

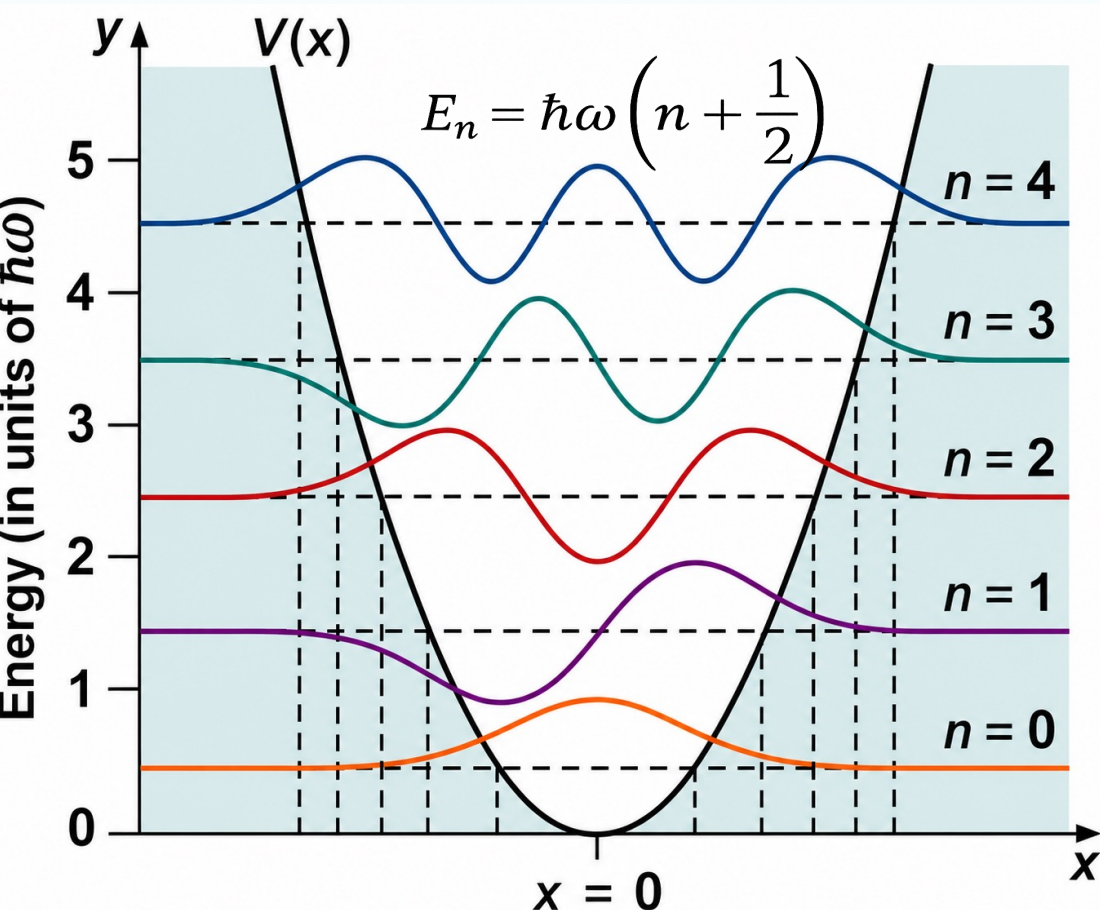
Qudit a d -dimensional quantum system

Fock level a quantum state with a set number of photons

SNAP gate applies phase shifts to individual fock levels

Disp gate mixes fock levels, changing the prob. distribution

Buffer levels additional fock levels beyond Target Unitary



A quantum harmonic oscillator can act as a qudit

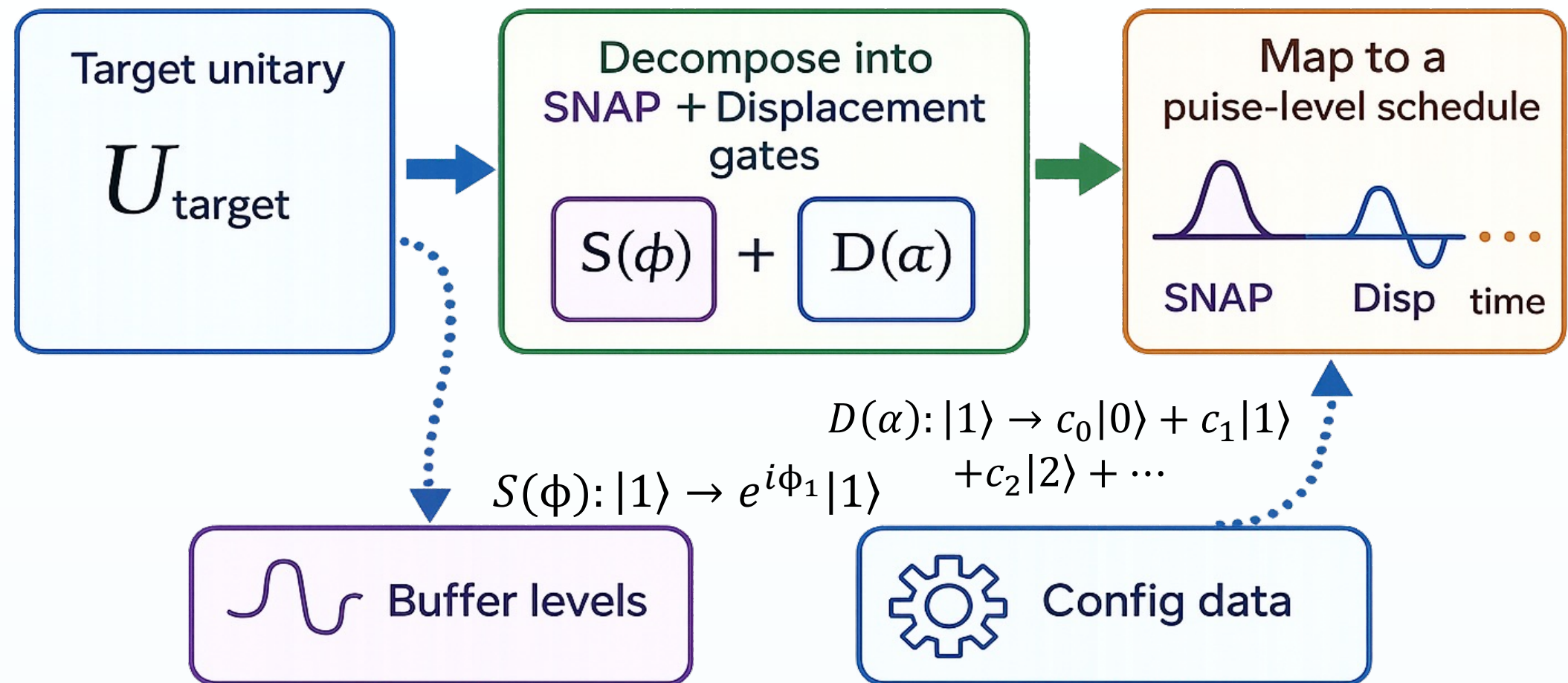
Future Research

1. Investigate alternative gate sets (e.g. ECD)
2. Study scalability at higher qudit dimensions
3. Develop multi-qudit decomposition methods
4. Design a hardware-aware decomposition

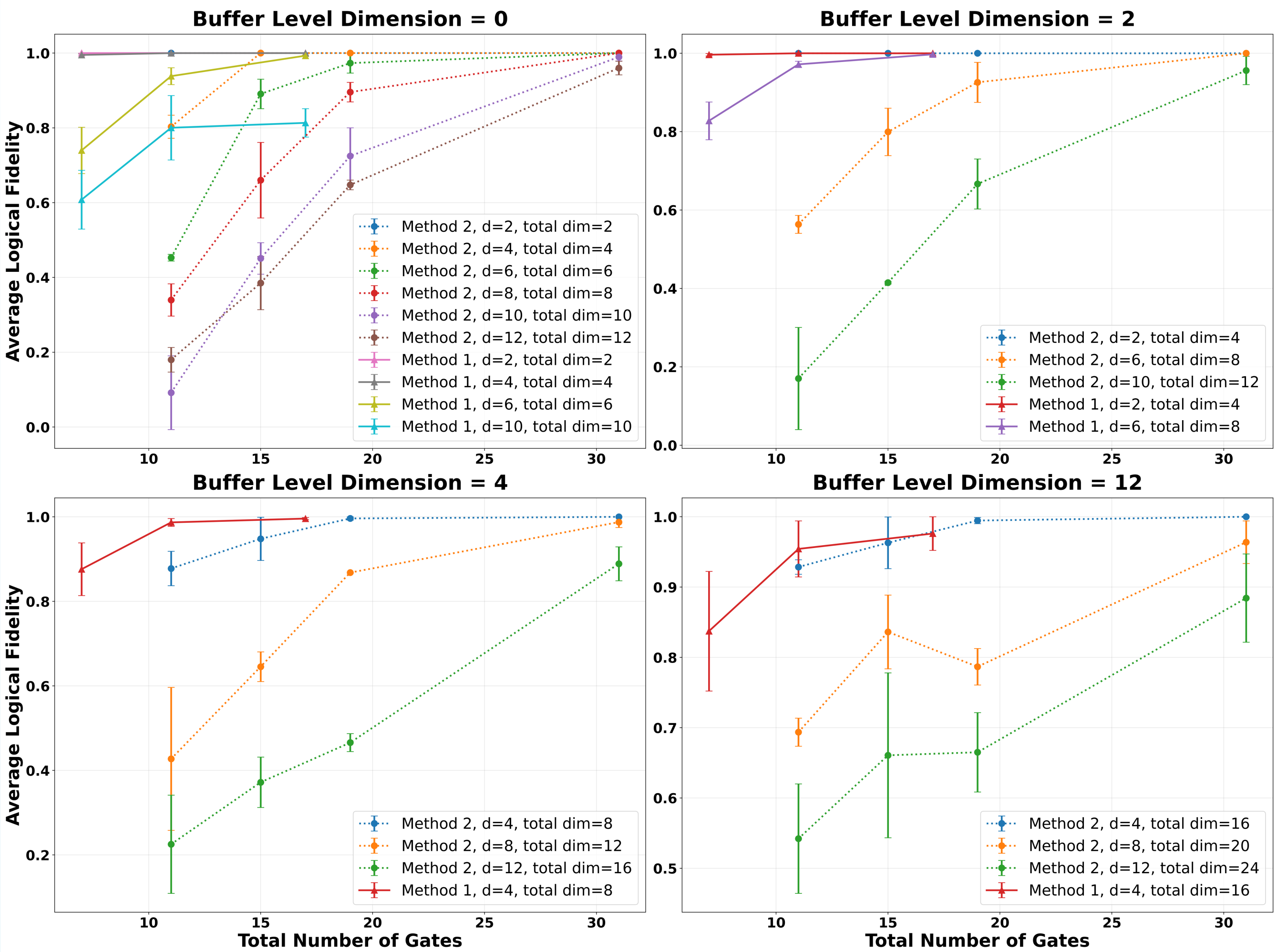
Acknowledgements & Refs

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Compilation Pipeline



Logical Fidelity vs Gate Count by Buffer Level Dimension



Method	d	b	Total Dim	# SNAP gates	#Disp gates	# total gates	fidelity
Method 1	4	4	8	8	9	17	0.999322
Method 2	4	4	8	15	16	31	1.000000
Method 1	6	2	8	8	9	17	0.999688
Method 2	6	2	8	15	16	31	1.000000

The dominate sources of noise are Photon loss (distorts probability) and Dephasing (destroys coherence).
Method 3 provides an exact approximation. However, it requires many more gates, allowing errors to accumulate and fidelity to degrade in the presence of noise

Methods of Decomposition

Method 1: A numerical synthesis method that decomposes a target unitary into a sequence of SNAP (Selective Number-dependent Arbitrary Phase) and Disp gates by optimizing all angle parameters simultaneously. The approach uses iterative optimization to minimize infidelity, often with multiple restarts to navigate a complex landscape. This method is flexible but can exhibit variability across runs

Method 2: A structured decomposition approach that builds gate sequences using guided heuristics or predefined patterns. Rather than optimizing the full parameter space at once, it leverages insight into the system to construct efficient circuits with fewer parameters. This can improve stability and interpretability compared to fully numerical methods

Method 3: An analytical decomposition method based on Givens-like rotations, where the target unitary is systematically broken down into elementary operations. Each step has a closed-form construction, leading to deterministic and reproducible circuits with lots of gates and high fidelity in a noiseless environment

Design Takeaways

1. **Adding buffer levels** does not always improve performance and can reduce fidelity
2. **Increasing gate count** improves noiseless accuracy but degrades under noise
3. **Decomposing larger unitaries** can be made easier by fixing angles, removing them from the computational space

Lindblad Noise Model

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_i (L_i \rho L_i^\dagger - 1/2\{L_i^\dagger L_i, \rho\}) \quad L = \sqrt{k\alpha}$$

Analytical Approach Method 3: Fidelity vs Total Gate Count

